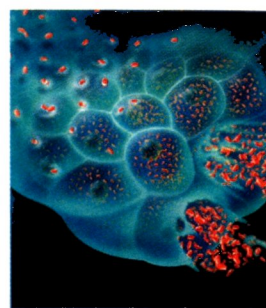
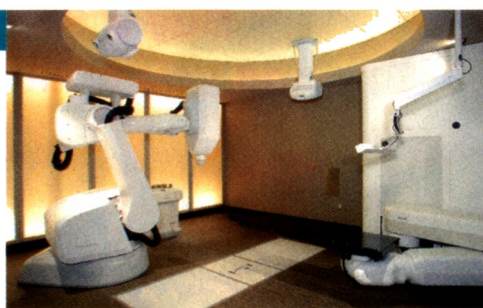


1 Sizing up the brain

Gene behind brain disorder found

This is what happens when the best brains come together to size up the brain. Geneticist Arun Kumar of the Indian Institute of Science (IISc) and psychiatrist Satish Girimaji of the National Institute of Mental Health and Neuro Sciences (NIMHANS) in Bangalore worked together for nine years. The outcome? Finding a key gene that causes microcephaly—a disorder marked by smaller-than-normal brain size and mental retardation. Until now only four genes were known to cause this hereditary disorder that develops in the foetal stage. This is the first time a fifth gene, named STIL, has been detected. The new gene is particularly valid for India, where one in every 50,000 to 1,00,000 live births ends up with microcephaly.

2



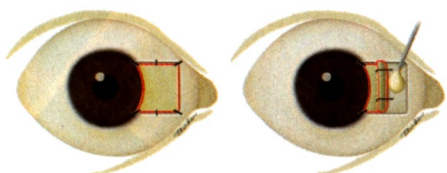
Beating cancer

Research and technology bring new hope

Medics link the HPV virus to cervical cancer, especially in sexually active women. In India, it kills about 76,000 each year. This year, biochemists, microbiologists and gynaecologists at the All India Institute of Medical Sciences sent a message of hope by detecting the most common HPV types that cause havoc in India. More good news: the Apollo Speciality Cancer Hospital in Chennai launched the CyberKnife technology, a first in India. A pocket-pincher, but it promises precise radiation to cancer cells, avoiding healthy tissue. A non-invasive option that caters to the quality of life of a cancer patient.

3 Gumming the eye

Using glue in eye surgery



For the first time, the entire front part of a patient's eye—cornea, sclera, iris, pupil and lens—was transplanted at Dr Agarwal's Eye Hospital & Eye Research Centre, Chennai. It followed the technique that was used here in 2007 to fix intraocular lens with fibrin glue (generally used to arrest bleeding and seal tissues in surgery) in a patient's eye where lenses could not be implanted by normal procedures. "Earlier, the treatment of damaged IOL was a challenge for ophthalmologists and in most cases nothing could be done," says Agarwal. "With this glue technology, we can now treat patients where intraocular lens capsules are missing."

BANDEEP SINGH/www.indiatodayimages.com



4 Lend me a hand, robot

Milestone robotic surgeries of chest and stomach

It was the year of surgeons using third-generation robots to reach a range of organs. The first such surgery on the thorax was done by Dr Arvind Kumar of AIIMS, Delhi, in June. Last month, Dr Jaydeep Palep did the first stomach surgery at Care Hospital, Hyderabad. "It's almost like shrinking one's hands and putting them in places they would never fit," says Kumar.

5 The sweet switch

A rare surgery for diabetes

Surgery for diabetes? That's exactly what Dr Surendra Ugale of Kirloskar Hospital in Hyderabad and Dr Ramen Goel of Bombay Hospital tried out through the Ileal Transposition (or small intestinal switch). The procedure shortens the intestinal tract between the stomach and terminal ileum, shifts it into an upper area and puts it in line again. The fallout? A biochemical process that facilitates insulin secretion in the presence of undigested food and controls Type II diabetes—a metabolic disorder that is marked by the failure to absorb sugar and starch due to lack of the hormone insulin.